Academium Engine Specification Packet

Section 1: Core Architecture & Orchestration

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1.1 Engine Identity Statement

The Academium engine is a **decision simulation engine that makes reasoning observable.**

Every behavioral rule, every architectural decision, and every output constraint in this specification exists to serve that single purpose. When a design decision conflicts with this statement, the statement wins.

The engine serves two users:

- **The student** — who practices decision-making in a safe, non-graded environment
- **The professor** — who defines the structure and uses the engine's outputs to enrich their teaching

The engine is not a tutor, grader, chatbot, or game. It is a structured environment that surfaces the consequences of decisions and makes the student's own reasoning visible to them.

1.2 Foundational Architecture Principles

These principles govern every implementation decision. They are not preferences — they are constraints.

1.2.1 Stateful World, Stateless Agents

All simulation state lives in the database. Agents are pure functions: they receive context as input and return structured output. No agent retains memory between calls. This means:

- Every agent call must receive the **full decision history** of the current session as part of its input context
- No consequence may be generated from a template that ignores prior decisions
- State continuity is the engine's responsibility, not the agent's

Why this matters (research basis): Kolb's Experiential Learning Theory requires that each new decision phase be informed by prior consequences. If agents don't receive history, the simulation becomes a series of disconnected episodes rather than a compounding learning trajectory. This destroys the pedagogical value.

1.2.2 Single Responsibility Per Agent

Each agent owns exactly one function. If an agent's responsibilities expand to two distinct functions, it must be split. This prevents monolithic agents that are hard to test, debug, and audit.

1.2.3 No Step May Be Skipped

The seven-step pipeline is executed in full for every student interaction. If any step fails, the pipeline halts and the student receives a graceful error message. A partial or malformed response is never delivered.

1.2.4 Explainability Over Sophistication

Every KPI movement, every narrative consequence, and every nudge must have a clear, traceable cause. If the engine cannot explain why something happened in plain language, the output is invalid regardless of how sophisticated the generation process was.

Why this matters (research basis): Kluger & DeNisi's Feedback Intervention Theory establishes that feedback only improves performance when it directs attention to the task and task-learning processes. Unexplained metric movements are not feedback — they are noise. They undermine trust and destroy the learning signal.

1.3 The Seven-Step Orchestration Pipeline

Each step is defined by: what it receives, what it must do, what it must output, what it must never do, and what happens if it fails.

STEP 1 — Input Reception

Purpose: Receive and validate the student's input before any processing begins.

Receives:

- Student input (MCQ option selection OR FR text string)
- Session state object (case ID, turn number, prior decision history, current decision prompt)
- Timestamp

Must do:

1. Confirm input type is either MCQ or FR
2. Confirm input is not null or empty (if empty → route to BLOCK logic in Step 2)
3. Record timestamp
4. Attach session state to the input object for downstream steps
5. Confirm turn number is valid (within the defined step count for this case)

Must output: A validated input object containing:

```
{
  input_type: "MCQ" | "FR",
  content: [string],
  turn_number: [integer],
  session_id: [string],
  timestamp: [ISO 8601],
  decision_history: [array of prior decisions and consequences],
  current_prompt: [string],
  case_id: [string],
  nudge_count: [integer — how many nudges have occurred at this decision point]
}
```

Must never do:

- Begin classification or interpretation of content
- Modify or clean the student's text in any way
- Make any judgment about the quality or relevance of the input

Failure mode: If session state cannot be retrieved or is corrupted → halt pipeline → deliver: *"We encountered a technical issue. Your progress has been saved. Please refresh and continue from where you left off."* Never deliver a partial response.

STEP 2 — Input Classification

Purpose: Classify the validated input as PASS, NUDGE, or BLOCK to determine how the pipeline proceeds.

Receives: The validated input object from Step 1.

Classification rules by input type:

For MCQ input:

- Always classify as **PASS**
- Rationale: All MCQ options are pre-authored as valid strategic postures. There is no invalid selection.
- Proceed immediately to Step 3.

For FR input — apply this decision tree in order:

BLOCK first (check these conditions before anything else): Classify as BLOCK if ANY of the following are true:

- Input is empty or contains only whitespace or punctuation
- Input consists entirely of random characters with no semantic content (e.g., "asdfghjkl", "123456")
- Input is clearly copy-pasted content with zero relevance to the case scenario
- Input contains profane, hostile, or offensive language

> **Critical rule:** Word count alone is NEVER a BLOCK criterion. A two-word response like "Priorizo clientes" is potentially a PASS. Brevity is not a disqualifier.

PASS next (if not BLOCK, check if any one of these is present): Classify as PASS if the input contains ANY of the following signals:

- A stated priority ("I would prioritize X")
- A reference to a specific case element ("Given the 15% budget cut...")
- An acknowledged trade-off ("This might slow delivery, but...")
- A named stakeholder impact ("The operations team would need to...")
- A reasoning chain, however brief ("X because Y")

> **Critical rule:** Grammar, spelling, and writing quality are NEVER classification criteria. The engine evaluates reasoning intent, not language polish.

NUDGE last (if not BLOCK and not PASS): Classify as NUDGE if the input shows engagement with the case but does not meet any PASS criterion:

- Relevant to the case but takes no clear position ("Es una situación difícil")
- A single potentially valid word without any justification ("Negociar")
- A question instead of a decision ("¿Qué pasaría si...?")

Nudge counter rule:

- Check the `nudge_count` field from the input object
- If `nudge_count` >= 2, override NUDGE → classify as PASS
- Rationale: nudges are advisory scaffolding, not gates. After two nudges, any submission proceeds.

Must output:

```
{
  classification: "PASS" | "NUDGE" | "BLOCK",
  classification_rationale: [string — brief internal note on why this
classification was assigned],
  proceed_to_step: 3 (PASS) | "NUDGE_RESPONSE" (NUDGE) | "BLOCK_RESPONSE" (BLOCK)
}
```

Must never do:

- Classify a brief but substantive response as BLOCK or NUDGE
- Factor in length, grammar, or language quality
- Generate any student-facing content at this step

On NUDGE — generate nudge response (before halting pipeline):

1. Preserve the student's text exactly as written — do not modify or clear it
2. Generate 1-2 clarifying questions using the mentorship posture
3. Display questions as a non-blocking callout adjacent to the input field
4. Increment `nudge_count` by 1
5. Halt pipeline — wait for student resubmission

Nudge language rules:

- Acknowledge the direction the student is taking ("Tu respuesta es válida...")
- Ask one or two specific clarifying questions
- Never suggest what the answer should be
- Never imply the response was wrong

Example acceptable nudges:

- "Tu respuesta es válida. Para hacerla más sólida, ¿qué estarías priorizando y por qué?"
- "Interesante dirección. ¿Qué stakeholders se verían más afectados por esta decisión?"
- "Buen punto de partida. ¿Qué estarías sacrificando al tomar este camino?"

On BLOCK — generate block response (before halting pipeline):

1. Preserve the student's text in the input field
2. Display a calm redirect message
3. Do not penalize, lecture, or express disappointment
4. Halt pipeline — student must resubmit

Block redirect templates:

- For profane/hostile input: "Entendemos tu frustración. Volvamos al caso — tu equipo necesita una decisión sobre [current situation]."
- For empty/nonsensical input: "Este simulador está diseñado para trabajar con el caso actual. ¿Qué decisión tomarías respecto a [current decision prompt]?"

Failure mode: If classification cannot be determined (edge case) → default to NUDGE, never to BLOCK.

STEP 3 — Signal Extraction

Purpose: Extract the reasoning signals embedded in the student's input that will drive consequence generation and metric computation.

Receives: The validated input object + PASS classification confirmation.

For MCQ input: Signals are not extracted from text — they are read from the pre-authored tradeoff signature attached to the selected option.

Tradeoff signature structure:

```
{
  option_id: [string],
  option_text: [string],
  strategic_posture: [string — e.g., "client-first", "cost-reduction",
"operational-efficiency"],
  kpi_impacts: {
    budget: [-2 | -1 | 0 | 1 | 2],
    team_morale: [-2 | -1 | 0 | 1 | 2],
    brand_reputation: [-2 | -1 | 0 | 1 | 2],
    operational_efficiency: [-2 | -1 | 0 | 1 | 2],
    stakeholder_trust: [-2 | -1 | 0 | 1 | 2]
  },
  narrative_branch: [string — which consequence template to use],
  tier_assignments: {
    budget: [1 | 2 | 3],
    team_morale: [1 | 2 | 3],
    ...
  }
}
```

For FR input: Extract the following five signals. For each signal, determine: present or absent, and if present, at what depth (surface or substantive).

Signal	Definition	Surface example	Substantive example
Intent	What the student is trying to achieve	"I want to help the client"	"I want to preserve the client relationship to protect the 30% revenue dependency"
Justification	Why the student believes this approach is right	"because it's important"	"because losing this client represents 30% of Q3 revenue"
Tradeoff Awareness	Acknowledgment of what is being sacrificed	"this might cause problems"	"this will delay internal projects by at least two weeks"
Stakeholder Awareness	Recognition of impact on specific groups	"the team will be affected"	"the operations team will need to reallocate two engineers"
Ethical Awareness	Identification of an ethical dimension	"we should be honest"	"we need to disclose the delay to the client before it affects their own planning"

Signal scoring:

- Absent: 0
- Surface (present but vague): 1
- Substantive (present with specificity): 2

Signal output object:

```
{
  intent: { present: bool, depth: 0|1|2, extracted_text: [string] },
  justification: { present: bool, depth: 0|1|2, extracted_text: [string] },
  tradeoff_awareness: { present: bool, depth: 0|1|2, extracted_text: [string] },
  stakeholder_awareness: { present: bool, depth: 0|1|2, extracted_text: [string]
},
  ethical_awareness: { present: bool, depth: 0|1|2, extracted_text: [string] },
  total_signal_score: [integer 0-10],
  dominant_signal: [string — which signal is strongest]
}
```

Must never do:

- Infer signals that are not present in the student's text
- Penalize absence of any single signal
- Use signal absence as a reason to downgrade consequence quality
- Extract from a template that ignores prior decision history

Failure mode: If signal extraction produces zero signals from a PASS-classified input → flag internally + treat as minimum signal threshold (total score = 1, intent present at surface level) → proceed. Never stall the pipeline on a PASS.

STEP 4 — Consequence Generation

Purpose: Generate the narrative consequence paragraph that describes what happened in the simulated organization as a result of the student's decision.

Receives:

- Signal extraction output (Step 3)
- Full decision history (from session state)
- Case consequence templates

- Turn number (to calibrate narrative complexity)

Generation rules — what the consequence MUST do:

1. Describe **observable outcomes** — what changed in the organization, what stakeholders said or did
2. Reference **specific elements from the case context** — maintain continuity with the scenario
3. Introduce **at least one new piece of information** the student did not have before (a stakeholder reaction, a market development, an internal discovery)
4. Reflect the **cumulative trajectory** — if this is Turn 2 or 3, the consequence must build on prior decisions, not exist in isolation
5. Reference the domain of **every KPI that will move** in Step 5 — metric movements must feel narratively supported, not arbitrary

Consequence length: 80-150 words. No more, no less. **Tone:** Professional, neutral, informative. The voice of a calm narrator describing business events.

The tone calibration test: Read the consequence aloud. If it sounds like a senior colleague describing what happened in a meeting, it passes. If it sounds like a teacher grading an assignment, it fails.

Prohibited language — these words and phrases may NEVER appear in consequence output:

Prohibited	Reason
"Correct" / "Incorrect"	Implies a right answer exists
"Wrong"	Same
"Best" / "Optimal"	Implies a superior path
"You should have..."	Retroactive judgment
"Unfortunately"	Loaded emotional framing
"Perfect"	Implies a ceiling was reached
"Mistake"	Moral judgment on a trade-off
"Well done" / "Good job"	Evaluative praise
"!" (exclamation point)	Performative tone

Permitted framing language: "This led to..." / "As a result..." / "Stakeholders responded by..." / "The team noticed..." / "This created pressure on..." / "An opportunity emerged..." / "The decision produced..." / "In response, [stakeholder] indicated..."

Signal-to-narrative mapping:

- High tradeoff awareness signal → consequence should surface both the positive and negative effects clearly
- High stakeholder awareness signal → consequence should reflect specific stakeholder reactions
- High ethical awareness signal → consequence should include an ethically-charged stakeholder response (not a lecture — a reaction)
- Low signal score overall → consequence is more neutral, surfaces less nuance, introduces a tension that the student did not anticipate

Post-generation validation (before proceeding to Step 5): Run a check for all prohibited language. If any prohibited term is detected → regenerate. This is not optional.

Must never do:

- State or imply what the correct decision was
- Express surprise or disappointment at the student's choice
- Generate a consequence that is identical or near-identical to a prior turn's consequence
- Generate from a template without incorporating the full decision history

Failure mode: If generation fails twice in succession → deliver a pre-authored fallback consequence from the case's consequence library (Turn-neutral, professionally toned). Log the failure. Never deliver a blank or error state to the student.

STEP 5 — Metric Computation

Purpose: Compute the KPI movements that result from the student's decision, and assign tier magnitudes.

Receives:

- Tradeoff signature (MCQ) OR signal extraction output (FR)
- Accumulated metric state (current KPI levels from prior turns)
- Case's pre-authored tier assignments (for Tier 3 movements)

Which KPIs move:

- A maximum of 3 KPIs may move per turn
- A minimum of 1 KPI must move per turn
- KPIs are selected based on the tradeoff signature (MCQ) or dominant signals (FR)

Tier assignment rules:

Tier	Label	Trigger Condition	Frequency
Tier 1	Small movement	Default for most decisions. Incremental impact.	60-70% of movements
Tier 2	Moderate movement	Clear strategic implication or named stakeholder impact	25-35% of movements
Tier 3	Large movement	Severe consequence. Ethical failure. Major pivot. MUST be pre-authored in case definition.	5-10% of movements

Critical anti-pattern rule — the tradeoff enforcement check: After computing all KPI directions, run this validation:

- Count how many KPIs are moving UP vs. DOWN
- If ALL KPIs are moving in the same direction → the decision has no trade-off → this is invalid
- Force at least one KPI to move opposite to the majority
- The KPI selected for reversal should be the one most logically connected to what was sacrificed in the decision

Rationale (research basis): The Tradeoff Posture (Engine Constitution §2.5) and the canonical case structure requirement that every decision involves genuine trade-offs. A consequence where everything improves or everything worsens communicates to the student that their choice was objectively good or objectively bad. This violates the non-graded principle and destroys the pedagogical value of trade-off learning.

No movement rule: Every turn must affect at least one KPI. A decision with zero metric impact signals to the student that their choice did not matter. This is never acceptable.

Must output:

```
{
  movements: [
    {
      kpi_id: "budget" | "team_morale" | "brand_reputation" |
"operational_efficiency" | "stakeholder_trust",
      direction: "up" | "down",
      tier: 1 | 2 | 3,
      delta_label: "small increase" | "moderate increase" | "significant increase"
| "small decrease" | "moderate decrease" | "significant decrease"
    }
  ],
  tradeoff_validation_passed: bool,
  accumulated_state: { [updated KPI levels after this turn] }
}
```

Must never do:

- Display a numerical score or percentage to the student
- Move a KPI without a corresponding causal explanation ready in Step 6
- Allow Tier 3 movements that were not pre-authored in the case definition

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STEP 6 — Causal Explanation Generation

Purpose: Generate the "Why did this change?" explanation for each KPI that moved in Step 5.

Receives:

- Metric movements output from Step 5
- Student's original input (the decision they made)
- Consequence narrative from Step 4
- Session state (full decision history)

Generation rules — one explanation per KPI movement:

Each explanation must:

1. Reference the student's **specific decision** (not a generic statement about the KPI)
2. Connect the decision to the KPI domain in **plain, professional language**
3. Be **1-3 sentences** — concise enough to read quickly, specific enough to be meaningful

4. Be **consistent with the consequence narrative** (no contradictions between the narrative and the causal explanation)

Quality test for each explanation: Replace the student's decision with a different decision. Does the explanation still make sense? If yes, the explanation is too generic and must be regenerated.

Example — acceptable: *"Your decision to expedite the client delivery without informing the operations team created unplanned overtime requirements, putting pressure on the budget allocation for Q3."*

Example — not acceptable (too generic): *"Budget decreased because the decision had financial implications."*

Must never do:

- State that the student's decision was wrong or suboptimal
- Use the explanation as an opportunity to teach a lesson ("This is why it's important to always communicate with stakeholders...")
- Generate an explanation that does not reference the student's specific choice

Must output:

```
{
  explanations: [
    {
      kpi_id: [string],
      explanation_text: [string — 1-3 sentences, student-specific, professional
tone]
    }
  ]
}
```

STEP 7 — Response Assembly

Purpose: Assemble the complete, validated response and deliver it to the presentation layer.

Receives:

- Consequence narrative (Step 4)
- Metric movements (Step 5)
- Causal explanations (Step 6)

- Session state (to determine if this is a mid-case or final turn)

Assembly structure — every response must contain these elements in this order:

```
1. CONSEQUENCE NARRATIVE
   [80-150 word paragraph describing what happened]

2. METRIC PANEL UPDATE
   For each KPI that moved:
   - KPI label (in case language)
   - Direction arrow (↑ or ↓)
   - Magnitude label (small / moderate / significant)
   - "Why did this change?" affordance (reveals causal explanation on tap/click)

3. NEXT STEP
   If mid-case: [Next decision prompt]
   If final turn: [Final outcome narrative — see Section 4 of this packet]
   If reflection turn: [Optional reflection prompt]
```

Pre-delivery validation checklist: Before any response reaches the student, confirm ALL of the following:

- [ ] Consequence narrative is 80-150 words
- [ ] No prohibited language appears anywhere in the response
- [ ] At least one KPI has moved
- [ ] At least one KPI has moved in the opposite direction of the majority (tradeoff check)
- [ ] Every KPI movement has a corresponding causal explanation
- [ ] The narrative references the domain of every KPI that moved
- [ ] The next decision prompt (or final outcome) is present
- [ ] No scoring, grading, or evaluative language appears anywhere
- [ ] The response is structurally complete (no missing fields)

If any item fails → do not deliver. Return to the relevant step and regenerate the failing element. Maximum 2 regeneration attempts per element before falling back to pre-authored content.

Must never do:

- Deliver a partial response (some elements missing)
- Deliver a response that failed the validation checklist
- Add any element not specified in the assembly structure
- Expose internal classification, signal scores, or pipeline metadata to the student

1.4 Pipeline Failure Handling

Failure Type	Response
Step fails once	Retry the step with the same inputs
Step fails twice	Use pre-authored fallback content for that step's output, log failure
Session state unrecoverable	Halt, display graceful recovery message, preserve all prior progress
Prohibited language detected post-assembly	Return to Step 4, regenerate consequence, re-run Steps 5-7
Complete pipeline failure	Display: "We encountered a technical issue. Your progress has been saved." Never show a partial or broken response.

1.5 Research Anchoring Summary

Every architectural decision in this section maps to a specific research foundation:

Design Decision	Research Basis
Full decision history passed to every agent	Kolb (1984) — experiential learning requires compounding experience, not isolated episodes
No step may be skipped	Sweller (1988) — cognitive load theory; partial or inconsistent outputs create extraneous load that impairs learning
Tradeoff enforcement (at least one KPI moves opposite)	Engine Constitution §2.5; Freeman et al. (2014) — active learning requires genuine engagement with complexity
Causal explanations mandatory for every KPI movement	Kluger & DeNisi (1996) — feedback must be task-specific and traceable to be effective
Nudge is non-blocking; max 2 per decision point	Vygotsky (1978) — scaffolding supports without replacing agency; ZPD requires the learner to complete the task
BLOCK only for empty/hostile/nonsensical input	Edmondson (1999) — psychological safety requires that students feel their effort is respected, not gatekept
Prohibited language filter	Bandura (1977) — self-efficacy is damaged by evaluative language; avoiding praise/criticism maintains the learning frame
Pre-authored Tier 3 movements only	Engine Constitution §5.3 — large metric movements must be pedagogically intentional, not algorithmically generated

End of Section 1. Next: Section 2 — Signal Extraction and Competency Mapping.